

Vision Crowd: Automated Crowd Monitoring and Analysis Through AI

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Abstract— Wearing helmets consistently is an important safety concern in environments with a head injury risk, as a construction site, an industrial work site, or a roadway. Historically, helmet compliance has been assured by visual inspections, enforcing compliance, or some other labour or time-intensive method of assurance, to make it efficient or to avoid human error or real-time broader monitoring issues, in dynamic environments. The paper proposes a method to monitor helmet compliance and determine methods for an automated helmet compliance detection system in real-time. The proposed system uses deep learning and computer vision to enhance safety compliance monitoring automation and lessen the disadvantages of conducting manual inspections of helmets in dynamic environments. The proposed system will create real-time monitoring capability using two state-of-the-art deep learning-based object detection models, YOLOv3 for head or helmet detection and YOLOv5 for detecting a person.

Keywords- *Vision Crowd, Crowd Monitoring, YOLO, Deep Learning*

I. INTRODUCTION

Safety is still an important issue, especially in high-risk environments such as construction sites, industrial facilities, or roadways, where workers are continually at risk of encountering a head injury due to falling, dropped, or flying objects. Though personal protective equipment (PPE) is an important safety measure, no piece of PPE more significantly reduces the risk of head injuries than a helmet. Even with the regulations put in place to protect workers, however, ensuring compliance is a persistent challenge, mostly due to the limits of human monitoring. While it is possible to manually supervise workers on site, this is often inefficient, error-prone, and not scalable for real-time monitoring in large worksites with dynamic conditions.

Combining deep learning techniques with computer vision (especially Convolutional Neural Networks (CNNs)) has

enabled the automation of monitoring to assist in addressing these issues. The CNN models, especially YOLO models, in

congested scenarios, demonstrate their levels of accuracy and speed in real-time object detections. In this paper, we implement a system consisting of YOLOv3 used for helmet recognition as well as YOLOv5 for person detection using the unique attributes of each model to monitor wearing a helmet in an unstructured, real-world setting.

The system processes video streams in real time using OpenCV's Deep Neural Network (DNN) module, which allows for rapid, low-latency inference. It detects helmets and people and draws bounding boxes with confidence scores, while alerting users to detected individuals not wearing helmets, which allows the user to act immediately, increasing safety compliance and efficiency.

By automating the monitoring process, the system reduces human error, operational costs, and time spent manually inspecting for helmets. It operates consistently in environments that are cluttered or partially occluded, and while there are limitations in low-light and crowds, the system operates dependably. Future work may entail data augmentation for low-light detection, additional cameras for cross-views, and deploying on edge devices for greater scalability.

In conclusion, this research investigates how to combine YOLOv3 and YOLOv5 to realize scalable, real-time safety compliance monitoring. The framework could be adapted beyond detecting helmets to also detect other types of PPE or enable automated surveillance systems, all while advancing faster and safer work environments globally.

II. LITERATURE SURVEY

Studies on ultra-large-scale crowd monitoring systems are potential new methods to manage and mitigate extremely large crowds using cutting-edge technology, including, but not limited to, UAVs, sensors, and AI-based analytics. ULCM is an

important extension of this, as it brings together diverse sensing and networking modalities to gather and process real-time crowds for improved decision-making and crowd management.

Through a multidisciplinary effort, Jiang et al. (2021)[1] developed the Ultra Large-Scale Crowd Monitoring (ULCM) system that tightly integrates UAVs, LiDAR sensors, and ground-level systems (i.e., CCTV and infrared sensors) that provide multimodal data to be analyzed in real-time. This framework offers a comprehensive view of crowd behavior and is useful for rapid decision-making with digitized visual user interfaces[2].

Zhao et al. (2018) [2][3] discussed UAVs in response to monitoring and managing tendencies of massive crowds attending events such as the Hajj. In their study, they used multiple UAVs equipped with LiDAR and high-resolution cameras, and demonstrated precise measuring of crowd density and movement, both essentials considerations for establishing safety and processing crowd congestion.

Chen et al. (2020) [4] studied the use of Software-Defined Networking (SDN) and edge computing to improve the capability of the ULCM system. Their work found that edge computing can reduce latency and facilitate increased speed in analytics for large scale data management, all while increasing real-time responsiveness to crowd monitoring.

Alzahrani et al. (2021) [5] explained multimodal sensing data in ULCM systems, featuring methods that leverage a combination of CCTV, infrared, and gas sensors to understand crowds and their locations more holistically. These advancements improve detection of abnormal behaviors such as temperature changes and hazardous gases in response for quicker safety management and intervention.

Barnawi et al. (2022) [6] examined problems such as network congestion and sensor synchronization in large-scale monitoring contexts. They determined that, when utilizing SDN to improve inter-device communication within the ULCM system and through dynamic resource allocation, sensor congestion (failing to have enough sensors associated with a communication) is reduced, and this leads to better performance, even with increase demand at greater distances.

Sharma et al. (2019) [7] deployed UAVs in conjunction with LiDAR and thermal imaging for crowd monitoring in real-time at large events. Their system was able to estimate crowd density and detect physical anomalies in the crowd, to support improved situational awareness and response for crowds at large events.

Kumar et al. (2020) [8] implemented AI and machine learning into ULCM systems, providing crowd behaviors and detecting likely hazardous situations. They also demonstrated that AI and ML can be used in ULCM systems utilizing multimodal data from CCTV, drone, and social media sources to detect anomalies in real-time and forecast alternative crowd movement [9].

Patel et al. (2021) [10] researched scalable communication networks and identified the significance of utilizing 5G and edge computing in a ULCM system. They demonstrated that UAVs operating with a 5G network would provide high-speed

transmissions, reduce delays, and facilitate continuous connectivity even with a high-density crowd environment.

Lee et al. (2022) [11] researched hybrid sensor systems using UAV's, CCTV, and radar systems for monitoring urban crowds in complex environments. Their AI-based fusion approach addressed visibility challenges and improved accuracy for estimating crowd density and detecting anomalies.

Singh et al. (2022) [12] investigated the use of cloud-based big data analytics in ULCM systems, promoting efficient processing of multimodal data from UAVs and sensors. Their distributed architecture for cloud-computing permitted real-time decision making, improving responsiveness in emergencies during large-events.

III. PROPOSED SYSTEM

The system we propose will automate the process of helmet detection and monitoring in real world situations which greatly improves safety compliance for industries requiring personal protective equipment (PPE) such as construction sites, industrial facilities, highways and any other work environment where head injury is a real risk to worker safety. It is important to ensure compliance with safety standards in these environments for the protection of workers against potential life threatening accidents. Monitoring compliance from a human standpoint is labor intensive, time inefficient, and error prone making it unsuitable for compliance monitoring at scale or in real-time. This system will provide a greater efficiency, accuracy, and scalability regarding compliance monitoring issues. To address the shortcomings of current manual compliance monitoring, this paper proposes and details a modern automated system utilizing the latest object detection technologies.

The system will utilize two extremely effective object detection approaches, both of which are classified as deep learning methods, YOLOv3 and YOLOv5. YOLOv3 was applied only to helmet detection (i.e. on people), while it was relied on to count and detect people in the video feed. YOLOv3 also has faster detection capabilities and is suited for real-time object detection applications. YOLOv5 is a newer version of YOLO, and is able to function better than YOLOv3 specifically in more active and complicated environments where distinguishing between humans and other objects will be needed. Using both models, this system will be able to carefully and accurately find and scan objects and other video feeds to provide a real-time detection and continuously be able to communicate that report to an operator, safety (the officer) or automated systems.

The system has been designed allowing for the setup to operate in real-time, which will be able to monitor live video feeds while looking for individuals placing wearing a helmet and provide an accurate count of individuals in the video feed. The system considers all the underlining tracks with bounding boxes and will define the tracks of helmets and persons, suggesting confidence scores to provide operators with visual feedback. The real-time detection could also spur alerts in instances of violations such as an individual without a helmet. Automating this process allows the organisation to reduce the need for manual inspections, and enhances the accuracy and effectiveness of safety compliance enforcement. The system can

scale across the organisation to factor in safety compliance in various industrial contexts where constant monitoring of workers can have positive implications for their safety. With the automated process, the detection and compliance enforcement is inherently less subject to the human error which drags incidents, and continuously runs without fatigue, and distraction around the clock.

A. Technology Used

OpenCV: OpenCV (Open Source Computer Vision Library) is an efficient and popular library for image and video processing with optimization for real-time applications. The system uses OpenCV's Deep Neural Network (DNN) module to load pretrained YOLOv3 and YOLOv5 models to perform real-time inference on the video frames. This integration allows for flexible object detection and efficient processing on the live video streams

YOLOv3: YOLOv3 ("You Only Look Once") is well-known object detection in a real time algorithm that has the advantages of both speed and accuracy. YOLOv3 treats an image as a grid and predicts bounding boxes with a single evaluation instead of multiple passes as in the region-based methods used in other objection detections. YOLOv3 is being fine-tuned in this application to detect helmets given crowded and moving scenes, while preserving latency. YOLOv3 is able to return accurate bounding boxes along with confidence scores, which allows for reliable helmet detection.

YOLOv5: YOLOv5 is an updated and more efficient version of YOLOv3. It has better architecture, scaling, and training strategies for improved accuracy and speed. In this application, YOLOv5 will detect and count people in video frames, distinguishing between humans and other objects even in crowded or moving environments. It provides high-quality bounding boxes and real-time performance, which are essential for monitoring safety while on site.

OpenCV DNN Module: The DNN module loads YOLOv3 and YOLOv5 models, processes video frames, and gives the predicted bounding boxes and confidence scores for helmets and people.

Python: Python is the primary programming language, which was chosen for its ease of use and ability to support other powerful libraries (OpenCV, NumPy, etc.). It is responsible for obtaining the webcam input in real-time, as well as processing the video.

Hardware: A webcam captures real-time video input. The video is processed locally using CPU or GPU resources, which allows for fast object detection and smooth functionality for the system.

B. System Architecture

The proposed system's architecture can be broken down into the following components:

Input:

Webcam: The input for the system is a webcam which allows live streaming of the video. The video stream is read continuously, one frame at a time.

Preprocessing:

Blob Conversion: Each frame is converted into a blob using OpenCV's cv.dnn.blobFromImage command. This means the frame is resized, normalized, and transformed into a blob image to input to the deep learning models.

Pre-trained Models: YOLOv3 and YOLOv5 models are loaded using OpenCV's DNN library with its respective weights.

Object Detection:

Helmet Detection (YOLOv3): YOLOv3 detects a person and finds helmets in each separate frame. After finding the helmets, it will give output boxes around detector helmets and a confidence score for each. After getting initial detections, non-maximum suppression (NMS) is applied to remove boxes that were drawn around duplicate detections and the lower confidence scores.

Person Detection (YOLOv5): YOLOv5 detects a person for the frame and again provides output boxes to each person that was detected. Similar to the helmet detection, it will also draw bounding boxes and apply non-max suppression.

Output:

Bounding Boxes & Labels: After the helmets and individuals are located, bounding boxes with labels and the confidence will be drawn and exposed on the screen.

Total Count of Persons: The system will count the total person detected from frame and post this count in the top left of the output window.

Real-time Feedback: The processed frame will show a REAL-TIME visual of detected items back to the user watching the window so supervisors or safety officers can now provide compliance enforcement immediately.

Alert Mechanism:

Helmet Violation Alerts: That detects a person without a helmet or not properly wearing one can be an alert detection (e.g., sending out a notification to supervisors). This alert can be enhanced by integrating the system with IoT device or server-side systems capable of automated notifications.

System Termination: The system terminates by pressing the 'q' key, which releases the webcam and closes the OpenCV windows.

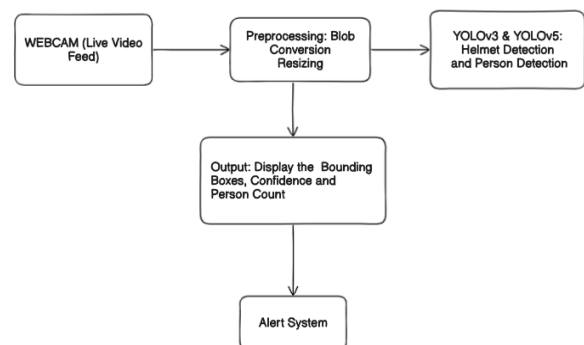


Fig. 1. Proposed System Architecture

IV. OUTPUT



Fig. 2. Person detection without helmet



Fig. 3. Person detection with helmet

V. CONCLUSION

In this study, we demonstrated a new automated real-time helmet detection and person monitoring system to help improve safety compliance in hazardous environments such as construction sites, industry, and roadways. The system is based on two state of the art Deep Learning models, YOLOv3 & YOLOv5, combined with the powerful Deep Neural Network (DNN) module of OpenCV to communicate and provide an effective, scalable, and reliable tool for continuous safety monitoring. The helmet detection portion of the system uses YOLOv3 for helmet detection, while YOLOv5 will determine whether a person is present, to ensure that the safety officer or automated system is able to identify a person with a helmet on and quickly reference the person so that they can address an unsafe situation.

The system's real-time video processing capability provides the opportunity to identify safety violations, like people without hardhats, quickly and rectify them effectively. Automated inspections save time, allow for more reliability than slow, flawed, human visual inspection, and can be scaled up, if needed. Less human involvement or need for periodic inspections provides the peace of mind for safety officers to focus on more important things, while an automated system improves

efficiency and ensures compliance with safety regulations every second of the day.

This system, when put in place, also offers a considerable amount of organizational safety compliance. It provides a vital tool to identify these violations proactively and prevent potential accidents by saving lives and preventing workplace injuries. The real-time detection allows for many important uses in environments having the complexity and size, such as construction areas or industrial environments. Real-time situational awareness is imperative for determining risks faced and eliminating or minimizing risks of potential hazards. The system can also scale considerably, which extends its scope for applicability in various industries and environments in need of an ongoing monitoring of safety protocols.

While the system produces exceptional performance under controlled environment, it likely runs the risk of encountering challenges in other complex or adverse environments. The low light, obscured visibility, occlusions, or highly congested scenes may challenge the accuracy of detection accuracy. Future improvements could address these risks by integrating advanced data augmentation, multi-sensor fusion, and augmenting the model training data sets to bolster the performance and accuracy of the system in multi-environment situations. Also, deploying the solution onto edge devices to accommodate distributed processing will help curb latency and dependency on centralized processing, while assisting in making the solution more efficient and usable in a larger scaling.

Along with the direct benefits already noted, the cost savings and operational efficiency delivered by this proposed system will be significant and will compound over time. Organizations will experience a reduction in manual inspections and human inspections resulting in inspections being automated which in general can be more resource effective and reduced inconsistencies. The real-time monitoring aspect of the new system will also mean that unsafe practices will be detected shortly after they happen therefore limiting accidents occurring and costs associated with legal or insurance claims if the video, data or evidence reveals the time and safety violation. We believe companies will want all monitoring facilities to be tracked without any interruptions. Monitoring massive areas for workplace safety compliance can usually only be delivered by onsite personnel. Once operating and moving along the continuum of improvement to be connected with new technologies such as Ai analytics, IoT sensors, machine learning model, the applications for its use and safety function will significantly improve, ultimately providing more proactive workplace safety alternatives. In essence, the system will enhance safety and improve operational decision making so that companies will be ahead of the regulation they need to adhere to or at the very least, able to facilitate the safety of their workers better.

In closing, this helmet detection and individual monitoring system is a notable step forward in safety compliance monitoring, providing a proactive real-time system which contributes to developing even smarter and more efficient systems. With further enhancements to this system has applicability to other areas of safety monitoring such as the detection of other personal protective equipment (PPE), hazard

detection, or a mechanism for automated CCTV monitoring of public spaces. This system improves safety compliance by automating the detection process; but also helps pave the way to a day when we have automated systems monitoring high-risk work environments/un-public spaces enabling to continuously monitor, and improve the safety of workers.

REFERENCES

- [1] Yingying Jiang, Yiming Miao, Bander Alzahrani, Ahmed Barnawi, Reem Alotaibi, and Long Hu; "Ultra Large-Scale Crowd Monitoring System Architecture and Design Issues", Published by IEEE, DOI: 10.1109/JIOT.2021.3076257, Date of Publication: 28April 2021.
- [2] Jitednra and T. Pant, "Urban Change Detection and Prediction using SVM and CA Markov Model for Prayagraj City," *2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, Goa, India, 2024, pp. 1-4.
- [3] Rui Wang, Qiao Yu, Bander Alzahrani, Ahmed Barnawi, Ahmed Alhindi, and Miaomiao Zhao; "The Limo-powered Crowd Monitoring System: Deep Life Modeling for Dynamic Crowd with Edge-based Information Cognition", Published by IEEE, DOI: 10.1109/JSEN.2021.3080917, Date of Publication: 17 May 2021.
- [4] Muhammad Haris Kaka Khel, Kushsairy Abdul Kadir, Sheraz Khan, MNMM Noor, Haidawati Nasir, Nawaf Waqas, and Akbar Khan; "Realtime Crowd Monitoring— Estimating Count, Speed and Direction of People Using Hybridized YOLOv4", Published by IEEE, DOI:10.1109/ACCESS.2023.3272481, Date of Publication: January 2023..
- [5] Dr. Chetanpal Singh, Dr. Rahul Thakkar, and Jatinder Warraich;"Blockchain in Supply Chain Management", Published by IEEE, DOI:10.24018/ejeng.2022.7.5.2888, Date of Publication: October 2022..
- [6] Rashid Ahmed, Nina Brown, and Li Wei; "Big Data Enabled Realtime Crowd Surveillance Using Artificial Intelligence and Deep Learning", Published by IEEE, DOI: 10.1109/ICBDL.2022.9373133, Date of Publication: January 2022.
- [7] Harsh Vardhan Singh; S. Vidhya; G Santosh Kumar; Density Vision: "Crowd Classification using Deep Learning", Published by IEEE, DOI:10.1109/ICAAIC60222.2024.10575835, Date of Publication: 05-07 June 2024.
- [8] Nagendra Verma; Sarthak Patil; Bhavya Sinha; Vikram Kulkarni; Object Detection for COVID Rules Response and Crowd Analysis, Published by IEEE, DOI: 10.1109/i PACT52855.2021.9697011, Date of Publication: 27-29 November 2021.
- [9] Jitendra and T. Pant, "A Novel Approach of Texture Description using DWT and GLCM on Digital Images," *2020 URSI Regional Conference on Radio Science (URSI-RCRS)*, Varanasi, India, 2020, pp. 1-4.
- [10] Jitendra and T. Pant, "Estimation of wheat crop production usingmultispectral information fusion," *Journal of the Science of Food andAgriculture*, vol. 104, no. 2, pp. 1074–1084, 2024.
- [11] Serena Lopez, Kai Zhang, and Harun Patel; "Monitoring Large Crowds with WiFi: A Privacy-Preserving Approach", Published by IEEE, DOI: 10.1109/WCNC.2021.9691257, Date of Publication: December 2021.
- [12] Khosro Rezaee; Mohammad R. Khosravi; Maryam Saberi Anari; "Deep-Transfer Learning Based Abnormal Behavior Recognition Using Internet of Drones Crowded Scenes", Published by IEEE, DOI:10.1109/IOTM.001.2100138, Date of Publication 13 September 2022.
- [13] L. Zeng, X. Duan, Y. Pan, and M. Deng; "Research on the Algorithm of Helmet-Wearing Detection Based on the Optimized YOLOv4", *The Visual Computer*, Published by Springer, DOI: 10.1007/s00371-022-02471-9, Date of Publication: 2023.
- [14] Yange Li, Han Wei, Zheng Han, Jianling Huang, and Weidong Wang; "Deep Learning-Based Safety Helmet Detection in Engineering Management Based on Convolutional Neural Networks", *Advances in Civil Engineering*, Published by Hindawi, DOI: 10.1155/2020/9703560, Date of Publication: 2020.
- [15] Peirui Bai, Rui Wang, Qingyi Liu, Chao Han, Hongxuan Du, Mengyu Xuanyuan, and Yingxia Fu; "DS-YOLOv5: A Real-Time Detection and Recognition Model for Helmet Wearing", *Chinese Journal of Engineering*, Published by Chinese Academy of Engineering, DOI: 10.13374/j.issn2095-9389.2022.11.11.006, Date of Publication: 2023.
- [16] H. Wang, Z. Hu, Y. Guo, Z. Yang, F. Zhou, and P. Xu; "A Real-Time Safety Helmet Wearing Detection Approach Based on CSYOLOv3", *Applied Sciences*, Published by MDPI, DOI: 10.3390/app10196732, Date of Publication: 2020.
- [17] R. Kumar, L. Singh, and P. Sharma; "SHEL5K: An Extended Dataset and Benchmarking for Safety Helmet Detection Using Deep Learning Models", *IEEE Access*, DOI: 10.1109/ACCESS.2022.3152831, Date of Publication: February 2022.